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Assignment number and name: Task 4 – Take an EBS Snapshot and Restore It to Simulate Disaster Recovery

Submission date: 25 October 2026

AWS Region Used: us-east-1(N. Virginia)

Objective

The objective of this task is to simulate disaster recovery using Amazon EBS snapshots. This involves creating test data on an attached EBS volume, creating a snapshot of the volume, restoring a new volume from the snapshot, attaching and mounting the restored volume to an EC2 instance, and verifying that the original data is successfully recovered. Finally, all additional EBS volumes, snapshots, and the EC2 instance are deleted to complete the cleanup process.

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1. Assignment Summary and AWS Resource IDs

- The test file was created on the original EBS data volume, the volume was snapshotted, and a new volume was created from the snapshot. The restored volume was mounted at /restored and the original file content was read successfully.

Resource	Identifier / value	Recorded details
EC2 instance	i-0f949d64c72c81ce2	Name: varun-ec2-snapshot; type t2.micro; Amazon Linux 2023
Original data volume	vol-0a4e19daf6d6ab300	Name: varun-ebs-data; 8 GiB; gp3; us-east-1a
Snapshot	snap-0f27e17dc154a3eb5	Description: my-backup-snapshot; 8 GiB source; status Completed
Restored volume	vol-0dddade8365b353b9	Name: varun-ebs-restored; 8 GiB; gp2; us-east-1a; created from snap-0f27e17dc154a3eb5
VPC	vpc-0ec1a47ec674c7582	Default VPC in us-east-1
Subnet	subnet-05ed4bbcc58ace97b	Name: varun-public-subnet; 172.31.0.0/24; us-east-1a
Security group	sg-0281c787c172ba1e2	Name: varun-ec2-sg

2. Console Steps Performed in Order

2.1 Region and networking

- Selected US East (N. Virginia), Region us-east-1.
- Verified that a Default VPC existed (VPC ID vpc-0ec1a47ec674c7582). No subnet was listed, so a subnet was created named varun-public-subnet in Availability Zone us-east-1a with CIDR 172.31.0.0/24.
- Enabled auto-assign public IPv4 for the subnet. The route table showed the local route 172.31.0.0/16 and the default route 0.0.0.0/0 to Internet Gateway igw-0b3603c3a1b7711ae.

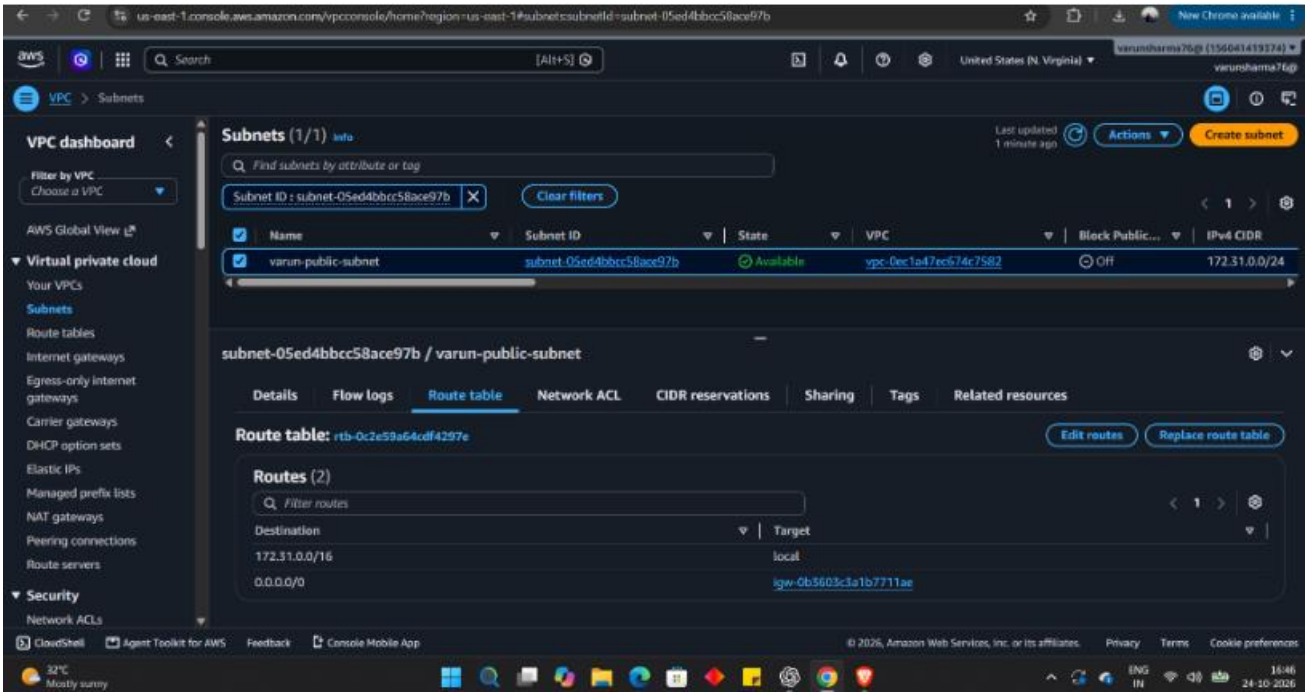


Figure 1. Subnet route table showing the default route through the Internet Gateway.

2.2 EC2 instance launch and connection

- Launched Amazon Linux 2023, 64-bit x86, instance type t2.micro, with instance name varun-ec2-snapshot.
- Configured the 8 GiB gp3 root volume and security group varun-ec2-sg. SSH was initially restricted to My IP; a second SSH rule was added using the Region-specific EC2 Instance Connect prefix list after the first browser connection attempt failed.

- Verified that the instance was Running, status checks were 2/2 passed, and the Availability Zone was us-east-1a. Connected using EC2 Instance Connect with username ec2-user.

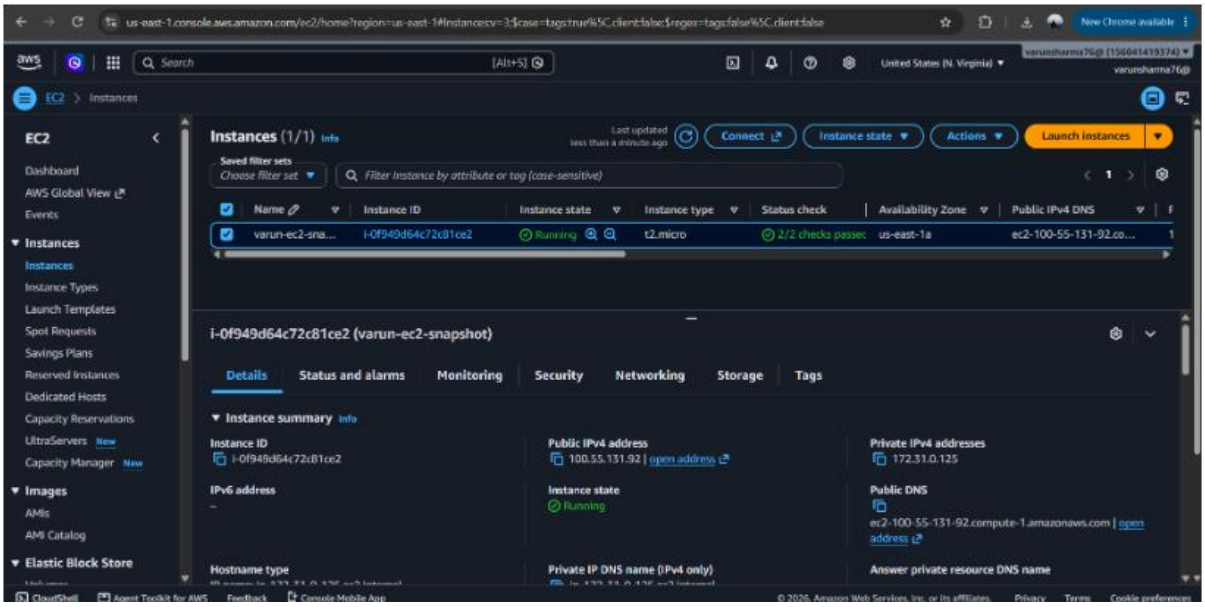


Figure 2. EC2 instance details: instance ID, Running state, instance type, Availability Zone, and account/Region context.

2.3 Create and attach the original data volume

- Created an 8 GiB gp3 EBS volume named varun-ebs-data in Availability Zone us-east-1a (Volume ID vol-0a4e19daf6d6ab300).
- Attached it to the EC2 instance using console device name /dev/sdg; the Linux device appeared as /dev/xvdg.

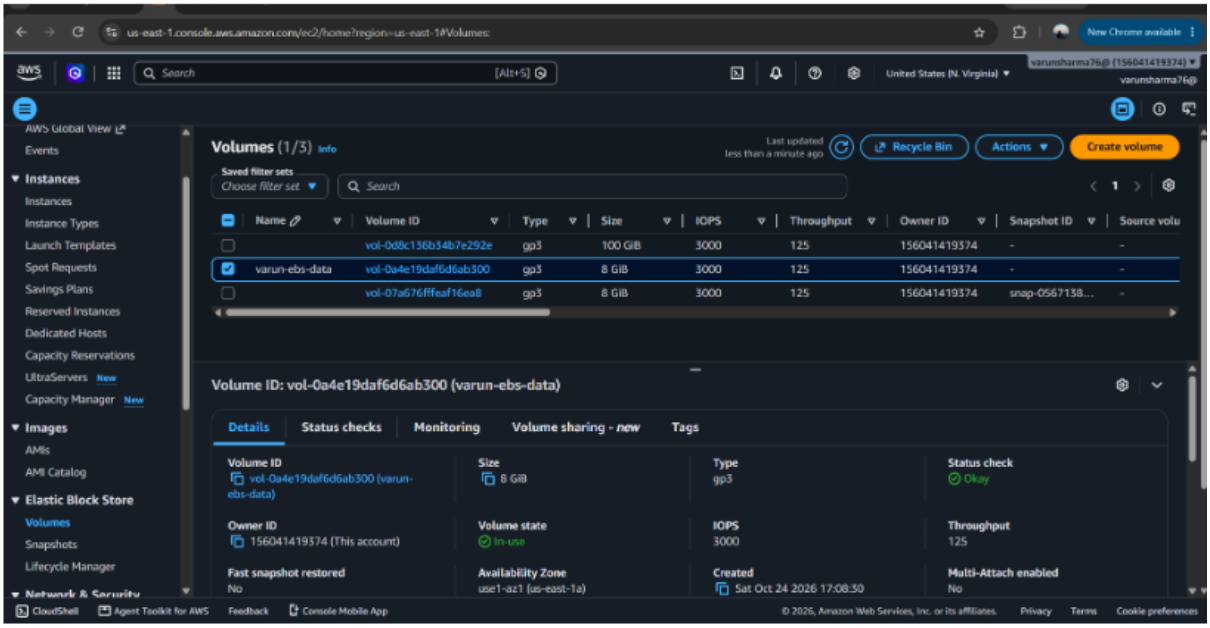


Figure 3. EBS Volumes page showing the original named data volume and volume IDs.

2.4 Prepare the original volume and create test data

The lsblk output identified /dev/xvdg as the separate 8 GiB data disk. The root disk was /dev/xvda and was not formatted.

```
lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS,TYPE
```

Output: The output listed /dev/xvda (8G root disk with partitions) and /dev/xvdg (8G data disk).

```
sudo file -s /dev/xvdg
```

Output: /dev/xvdg: data

```
sudo mkfs.xfs /dev/xvdg
```

Output: XFS filesystem creation completed successfully; the output reported metadata, data, naming, log and realtime sections.

```
sudo mkdir -p /mydata
```

Output: No output; the directory was created.

```
sudo mount /dev/xvdg /mydata
```

Output: No output; the volume was mounted.

```
findmnt /mydata
```

Output: /mydata /dev/xvdg xfs rw,relatime, ...

```
echo "Snapshot test data - Varun Sharma" | sudo tee /mydata/snapshot-test.txt
```

Output: Snapshot test data - Varun Sharma

```
cat /mydata/snapshot-test.txt
```

Output: Snapshot test data - Varun Sharma

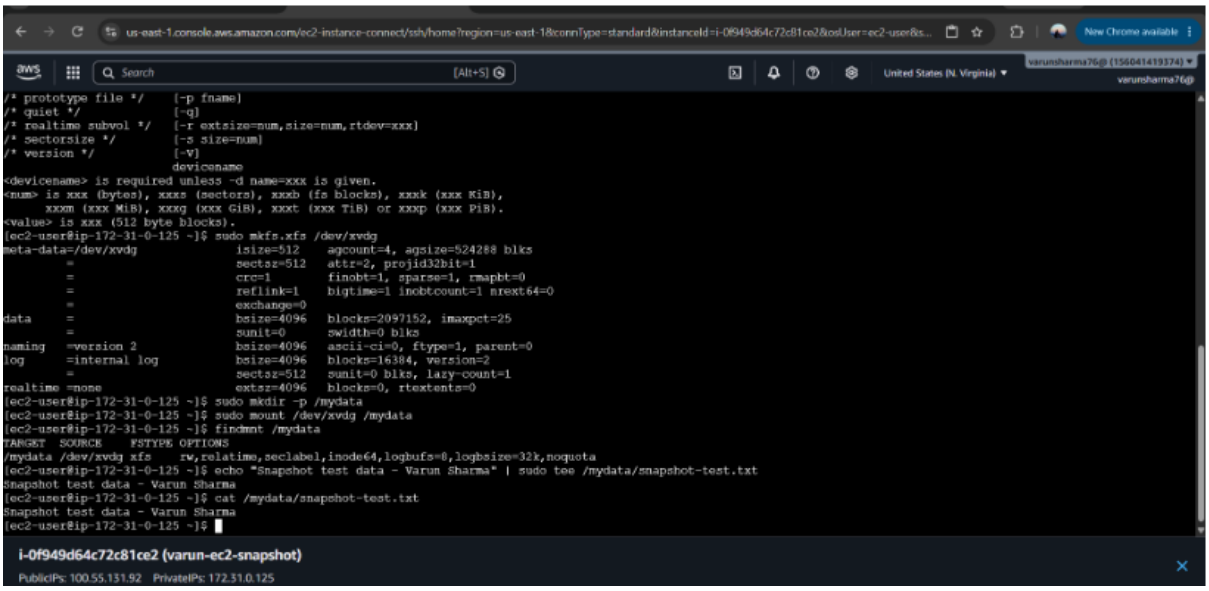


Figure 4. Original volume preparation, mount verification, test-file creation and cat output in the EC2 terminal

2.5 Create the EBS snapshot

- In EC2 → Volumes, selected varun-ebs-data → Actions → Create snapshot.
- Set the description to my-backup-snapshot and created the snapshot.
- Opened EC2 → Snapshots and verified the snapshot status was Completed. The Pending state was not captured in the screenshots shared here because the screenshot available already showed Completed.

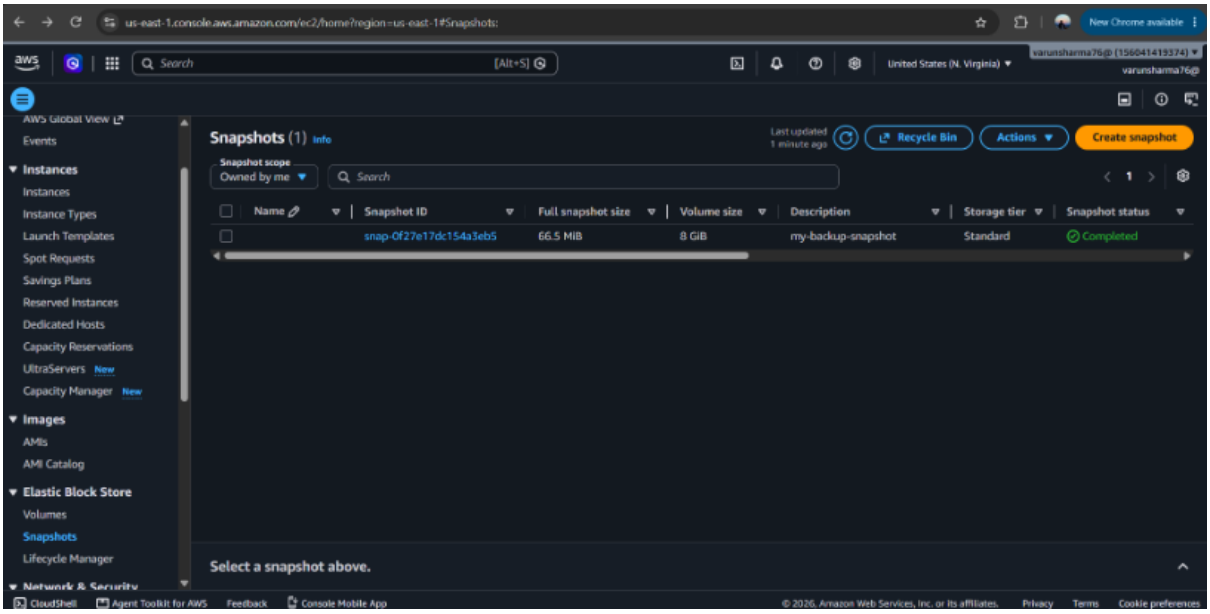


Figure 5. Snapshot snap-0f27e17dc154a3eb5, description my-backup-snapshot, volume size 8 GiB, status Completed.

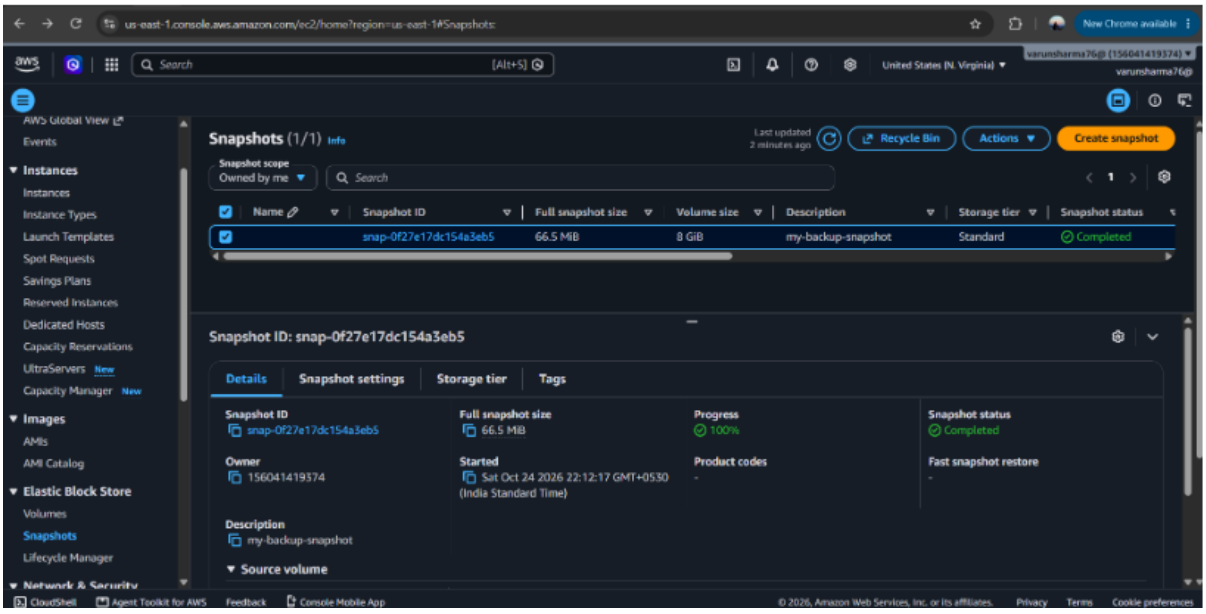


Figure 6. Snapshot details page showing snap-0f27e17dc154a3eb5, description my-backup-snapshot and Completed status.

2.6 Create the restored volume and attach it

- Created a volume from snapshot snap-0f27e17dc154a3eb5 with type gp2, size 8 GiB, Availability Zone us-east-1a and name varun-ebs-restored.
- Unmounted the original data volume from /mydata and detached the original volume before attaching the restored volume, so that the device name /dev/xvdg could be used for the restored volume.
- Attached varun-ebs-restored to instance i-0f949d64c72c81ce2 using console device name /dev/sdg (seen by Linux as /dev/xvdg).

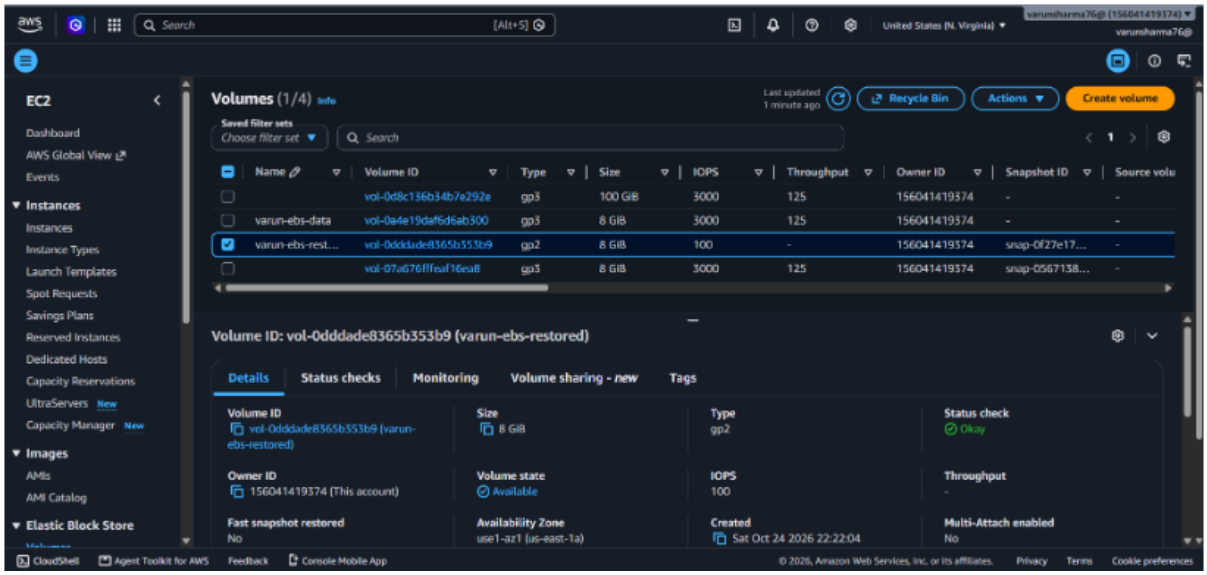


Figure 7. Restored volume varun-ebs-restored in Available state, created from snapshot snap-0f27e17dc154a3eb5; 8 GiB gp2 in us-east-1a.

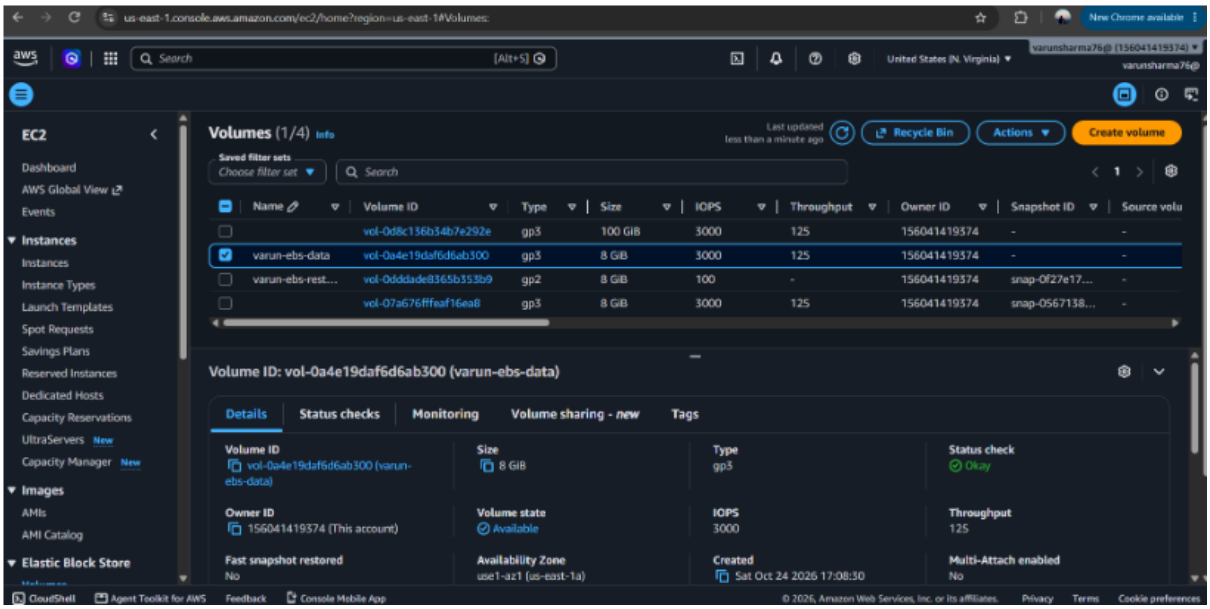


Figure 8. Original data volume in Available state after unmount/detach, allowing the restored disk to reuse the /dev/xvdg device name.

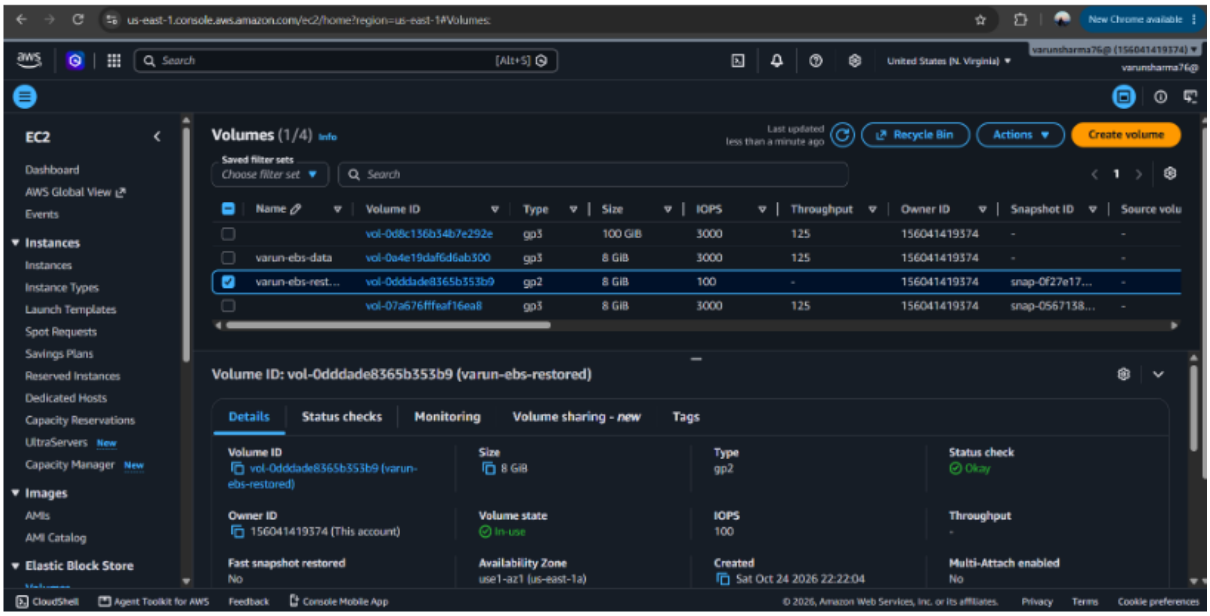


Figure 9. Restored volume attached to the EC2 instance; console state is In-use.

2.7 Mount the restored volume and verify the recovered data

```
lsblk -f
```

Output: The output listed /dev/xvdg with FSTYPE xfs and UUID d4b95390-6e0a-4b79-ae83-27a0151f6884; /dev/xvda remained the root device.

```
sudo mkdir -p /restored
```

Output: No output; the restore mount directory was created.

```
sudo mount /dev/xvdg /restored
```

Output: No output on the successful mount attempt.

```
findmnt /restored
```

Output: `/restored /dev/xvdg xfs rw,relatime, ...`

```
cat /restored/snapshot-test.txt
```

Output: `Snapshot test data - Varun Sharma`

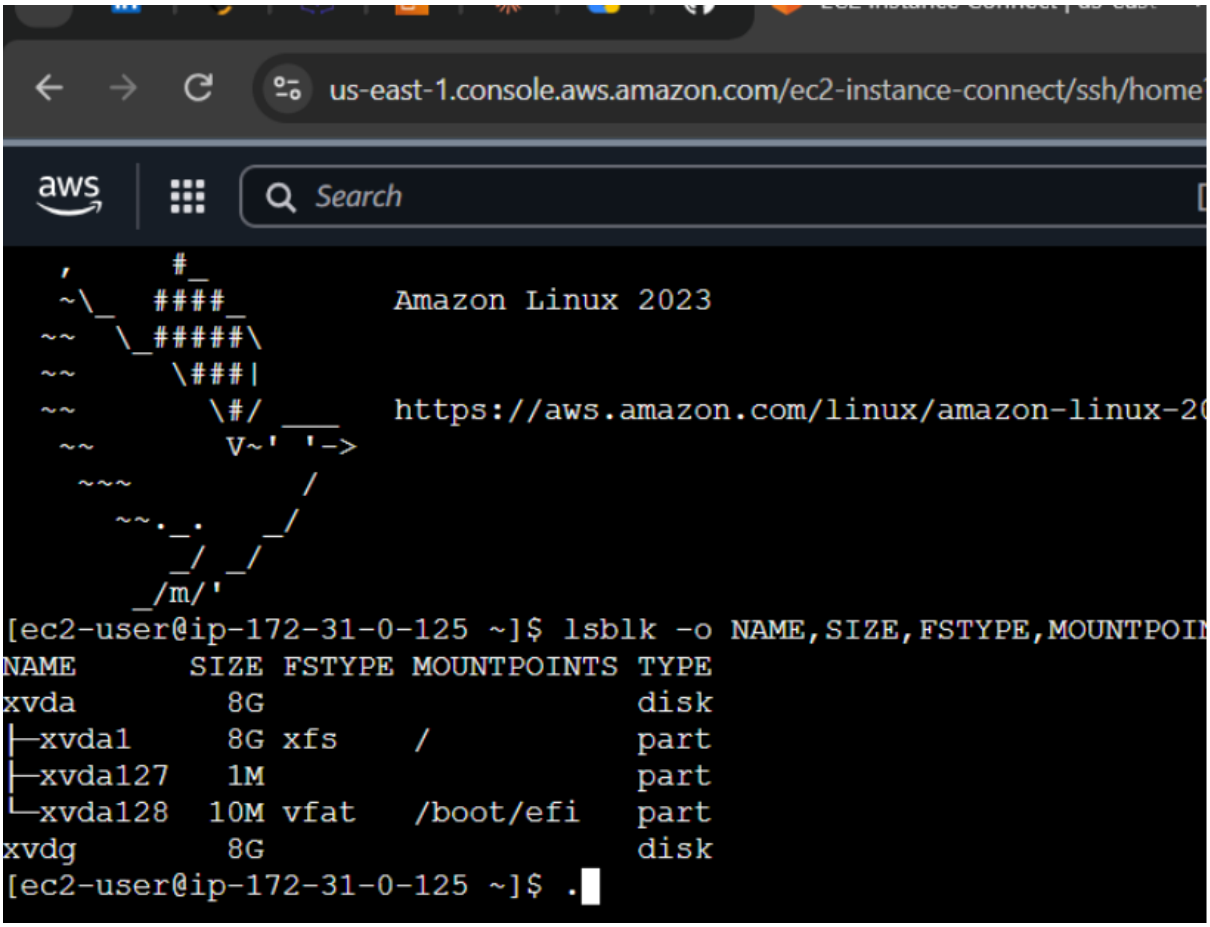


Figure 10. Terminal lsblk -f output after attaching the restored volume; /dev/xvdg is an XFS filesystem and the root disk remains /dev/xvda.

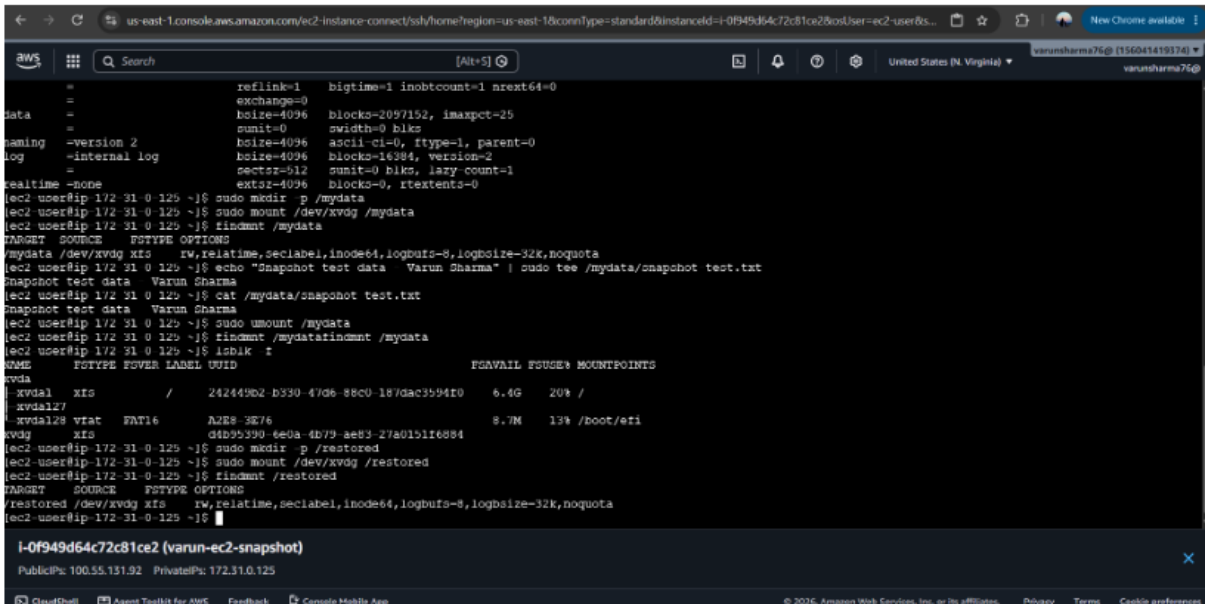


Figure 11. Terminal evidence that /restored is mounted from /dev/xvdg with an XFS filesystem.

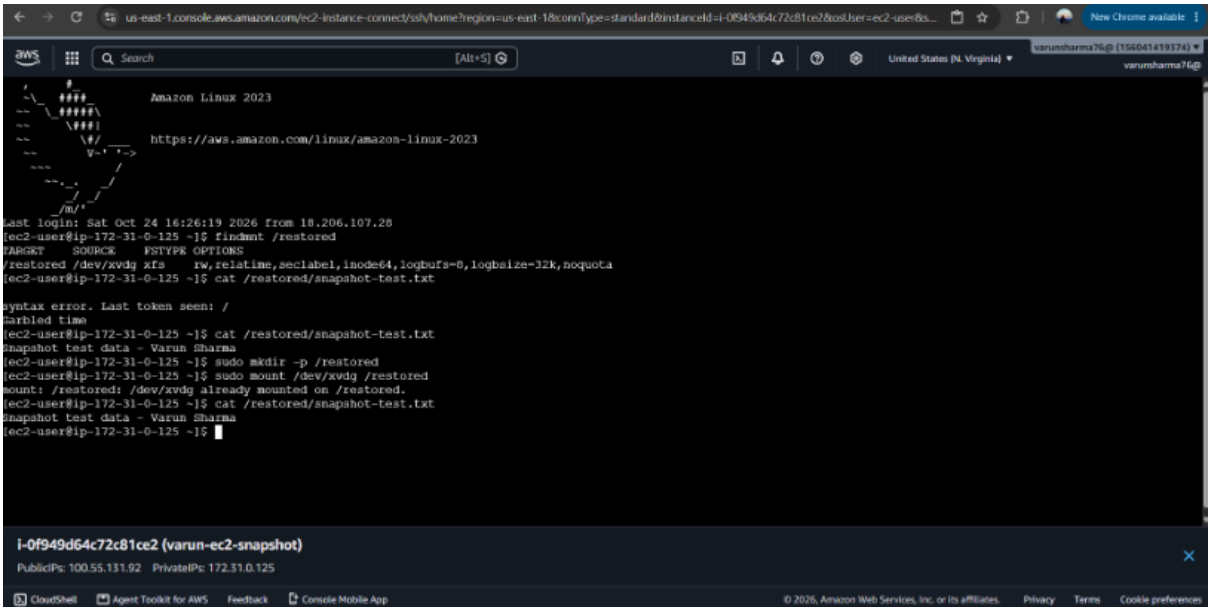


Figure 12. Terminal output showing the recovered file content: Snapshot test data - Varun Sharma. Earlier failed terminal input is also visible; the successful cat output is shown near the bottom.

2.8 Cleanup actions reported and evidence status

- Unmounted the restored volume with sudo umount /restored, then detached the restored volume in EC2 → Volumes; the supplied detached-state screenshot shows the restored volume as Available.
- The user reported deleting the restored volume varun-ebs-restored.
- The user reported deleting the original data volume varun-ebs-data.
- The user reported deleting snapshot snap-0f27e17dc154a3eb5; the available screenshot records the confirmation dialog, not the post-delete list.
- The user reported terminating EC2 instance i-0f949d64c72c81ce2; the supplied instance detail image is an earlier screenshot showing Running.

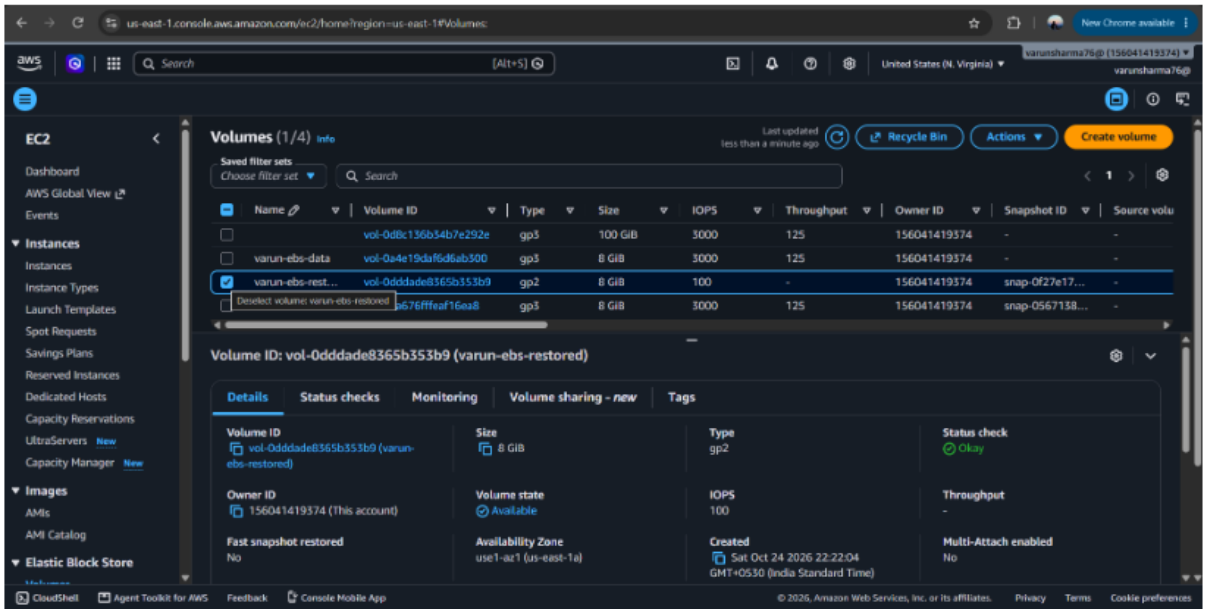


Figure 13. Restored volume in Available state after it was detached, prior to deletion.

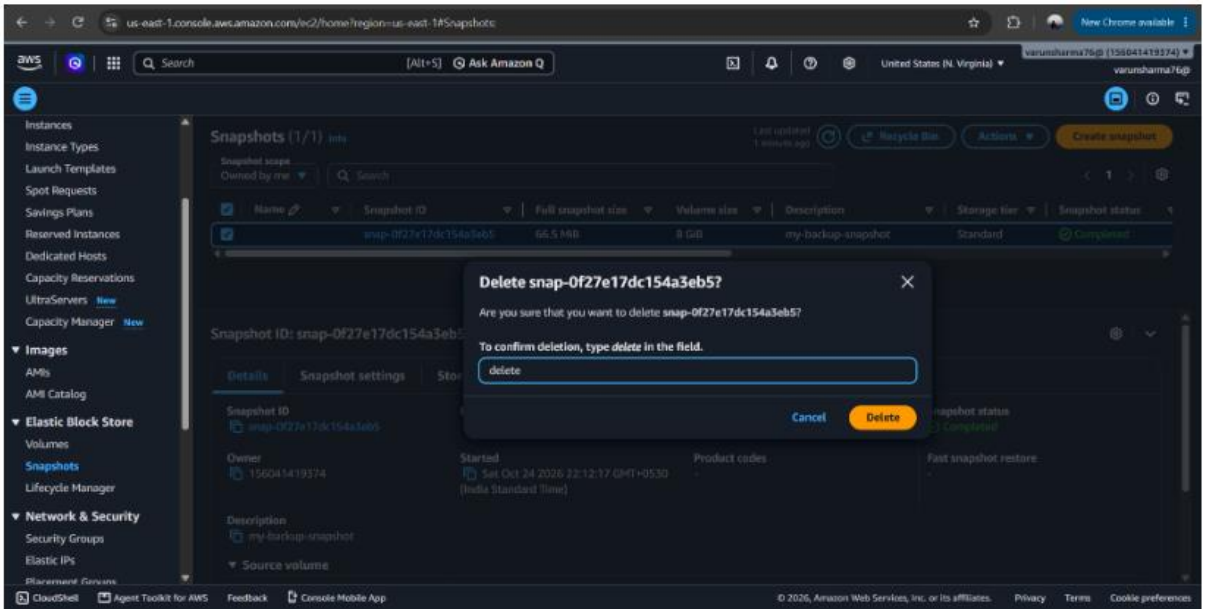


Figure 14. Snapshot delete-confirmation dialog. This records the delete-confirmation step; by itself it is not a post-deletion proof.